

MATHS TOP tips



The secret to becoming a **genius** at maths is to use shortcuts. All maths experts do this, from brilliant scientists to quick-thinking tradesmen. Some of the best tricks are shown on this page. When you've practised using them, you can carry out complicated sums in your head without even touching a calculator.

ROUNDING OFF

ADDING TOGETHER LARGE NUMBERS IN YOUR HEAD IS OFTEN EASIER IF YOU ROUND OFF ONE OF THE NUMBERS TO THE NEAREST 10. For instance, to add 46 and 39, round off the 39 to 40 by adding 1.

So, $46 + 40 = 86$. To finish off, subtract the 1 you added, to make 85.

STAIRWAY TO ELEVEN

Multiplying by 11 is easy if you remember that 11 is ten plus one. To multiply 63 by 11, simply multiply by 10 (to give you 630) and add 63 once, giving you 693.

HIGH FIVES

DIVIDING OR MULTIPLYING BY 5 IS MUCH EASIER IF YOU REMEMBER THAT FIVE IS HALF OF TEN. For example, to work out 5×36 , first work out 10×36 , which is 360. Then halve this for the final answer: 180. To divide a large number by 5, divide it by ten first and then double the answer. So, to find out $325 \div 5$, work out $325 \div 10 = 32.5$. DOUBLING THIS GIVES YOU 65.